

Outdoor WiFi + Camera Installation

Sand Volleyball Courts - Design & Placement Report

Manor Lanes / WNY Social Sports - 150 Grand Island Blvd

Site	3 outdoor sand volleyball courts + 2 patios (P1, P2)
Platform	Ubiquiti UniFi (Network + Protect)
Installer	Brian Russo (owner-installer)
Controller / NVR	CloudKey+ (SSD) + Network Video Recorder Pro + 16 TB HDD (existing)
Key constraints	No trenching; no new cable across sand/drainage; cold-weather (Buffalo NY)
Backhaul method	PoE-powered point-to-point wireless bridges (no outdoor cable runs)
Net new hardware	approx. \$1,400 - \$1,900 (most gear repositioned, not repurchased)

Design summary

A PoE-powered wireless bridge on the building beams a wired-equivalent link to a new 20 ft center pole between Courts 1 and 2 (the distribution hub). A second bridge relays from the center pole to the Court 3 pole. All cameras, an OPTIONAL action PTZ, and access points run wired-locally off small switches at each pole. The three existing U6 Mesh Pro APs are RELOCATED from the concrete building wall (the cause of the failed heat-map test) onto field poles, where their coverage radiates over the sand. No cable crosses the courts or drainage.

1. Background & Existing Conditions

Problem identified

- Two U6 Mesh Pro APs were wall-mounted on the concrete back of the building.
- Heat-map survey showed strong signal (-30 to -50 dBm) only at the wall; the entire court surface measured -70 to -80 dBm (marginal-to-unusable), failing under league crowds.
- Root cause: omnidirectional APs flat on concrete radiate left/right along the wall and reflect/absorb into the concrete - signal does not project across the open sand.

Fix

- Relocate APs off the wall onto field poles so coverage radiates down/out over the courts.
- Use directional, aimed, PoE-powered wireless BRIDGES (not omni APs) for backhaul - the concrete-wall limitation does not apply to a point-to-point link mounted high and aimed.

Existing UniFi equipment to be REUSED (no purchase)

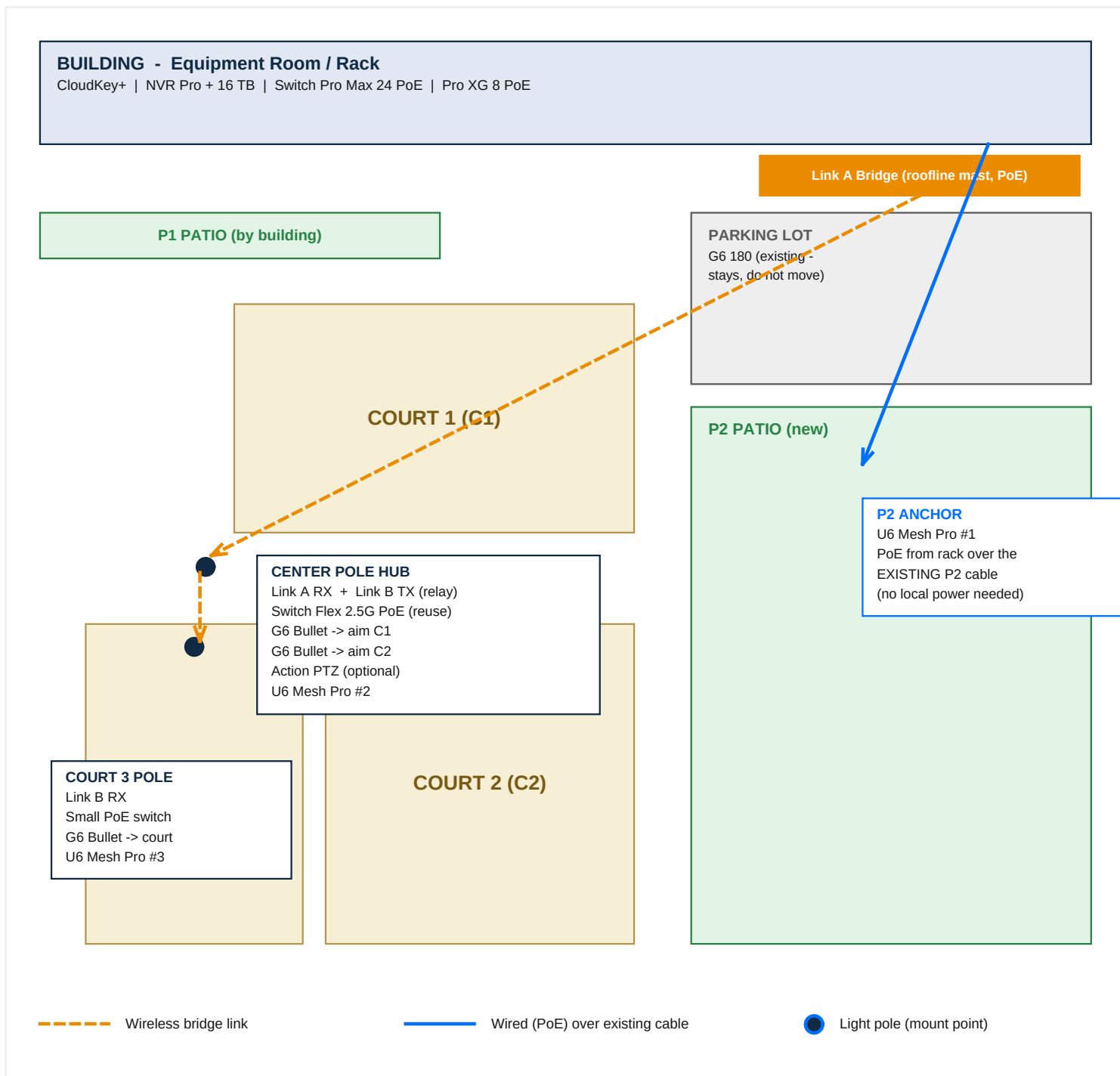
Item	Qty	Role in this project
Network Video Recorder Pro + 16 TB HDD	1	Records all court cameras
CloudKey+ (SSD)	1	UniFi Network/Protect controller
Switch Pro Max 24 PoE / Pro XG 8 PoE	2 / 1	Rack PoE source for building bridge & P2 AP
Switch Flex 2.5G PoE + 210W adapter	1	CENTER POLE local switch
Access Point U6 Mesh Pro	3	Relocated to P2, center pole, Court 3
AI PTZ Industrial camera (IP66, -40C)	1	STAYS on parking lot (not moved)
PoE++ 60W injector (spare)	1	Backup power for action PTZ if fitted
Spare G6 / G5 PTZ (if available)	?	Optional action cam - check fleet for a spare
Camera G6 180	1	STAYS on parking lot (do NOT use for courts)

Confirmed site facts

- Center pole: new 20 ft pole (2 lights) between Court 1 and Court 2 - POWER PRESENT.
- Court 3 pole: 20 ft light pole - POWER PRESENT.
- P2 patio post: existing Cat6 pulled to it, but NO power (AP here is PoE-fed from rack).
- Rack is inside the building, ~130 ft from the center-pole location.
- Courts each ~29'6" x 59'1" (approx 30 x 60 ft); perimeter light poles ~20 ft.

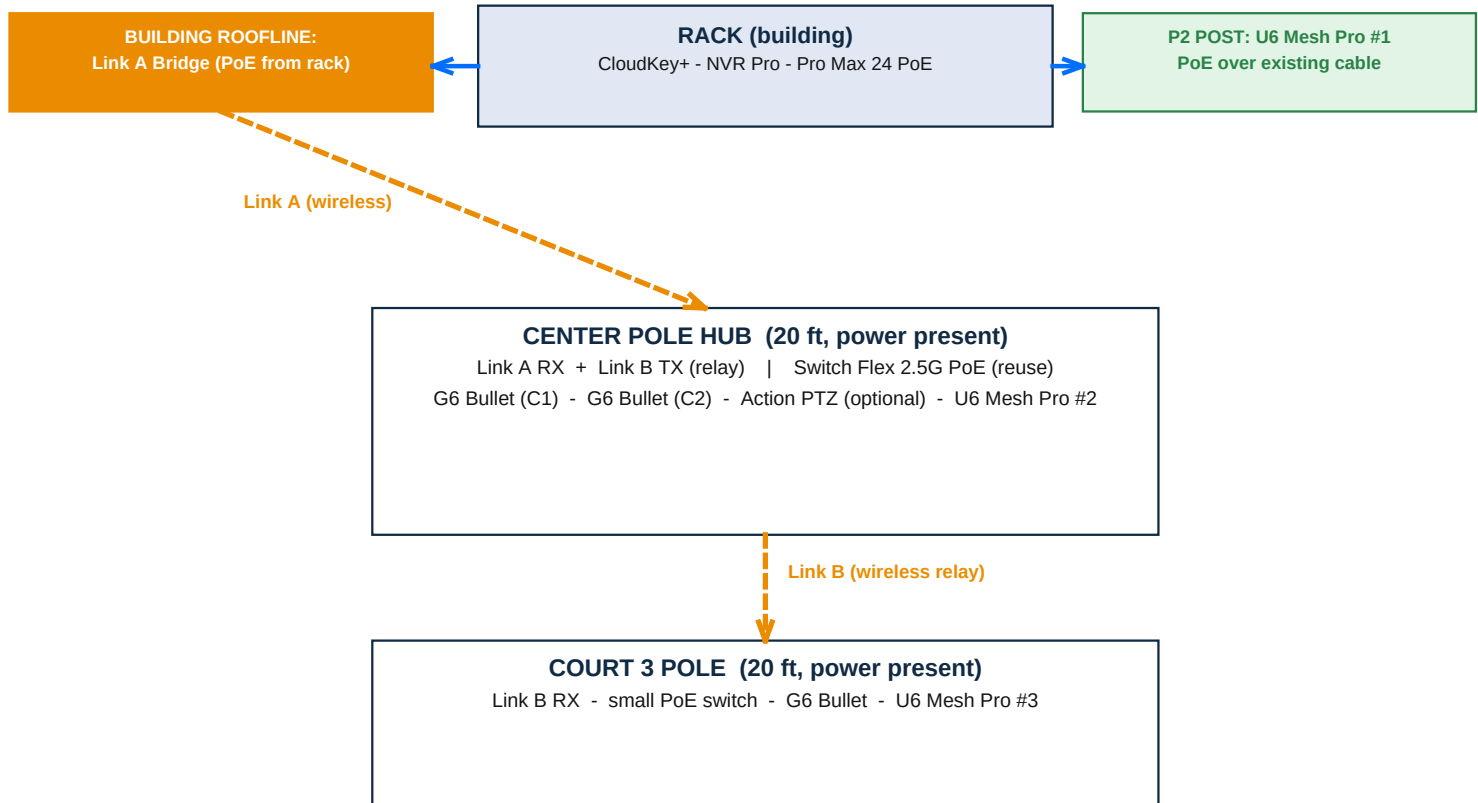
2. Site Layout & Device Placement

Schematic - relative positions per site photos/CAD; not to exact scale. Solid = wired (PoE). Dashed = wireless bridge.



Orientation per owner: building at top with rack; P1 patio adjacent to building; Court 1 nearer building, Court 2 further; Court 3 to the left behind Court 2; P2 (new patio) along the right with the existing pulled cable.

3. Network Topology & Wireless Links



Wireless bridge details

- Link A: building roofline -> center pole. Short (~130 ft), clear line-of-sight over courts.
- Link B: center pole -> Court 3 pole (relay). Court 3 sits behind Court 2 - no clean LoS from the building, so it relays through the elevated center pole.
- Hardware options: UniFi Building Bridge (UBB, 60 GHz, plug-and-play) OR 5 GHz airMAX (LiteBeam/NanoStation) for maximum winter margin. Either handles the load with ease.
- Throughput needed per link: ~50-80 Mbps (3x 4K cameras + AP clients). UBB delivers 1+ Gbps; airMAX 5 GHz delivers 300-450 Mbps. Large headroom.
- All bridge radios are POWERED OVER ETHERNET - no AC outlet needed at any bridge end.
- Mount the building bridge on a short mast ABOVE the concrete wall/parapet so the lip does not block the beam; aim at the center-pole radio.

4. Device-by-Device Placement & Mounting

Location	Device	Placement / Aim	Height	Power & Data
Building roofline	Link A Bridge (UBB/airMAX)	Short mast above parapet; aim at center pole	Roofline	PoE from rack switch
P2 patio post	U6 Mesh Pro #1	Top of post; radiate over P2 + east side	~12-15 ft	PoE from rack over EXISTING cable
Center pole	Link A Bridge RX	Aim back at building roofline	~18-20 ft	PoE from Flex 2.5G switch
Center pole	Link B Bridge TX	Aim at Court 3 pole	~18-20 ft	PoE from Flex 2.5G switch
Center pole	Switch Flex 2.5G PoE (reuse)	In NEMA enclosure on pole	~6 ft (reach)	AC from pole light circuit + 210W adapter
Center pole	G6 Bullet (Court 1)	Aim down length of Court 1	~18-20 ft	PoE from Flex switch
Center pole	G6 Bullet (Court 2)	Aim down length of Court 2	~18-20 ft	PoE from Flex switch
Center pole	Action PTZ (OPTIONAL, special games)	Roaming/zoom over C1+C2; see BOM choices	~18-20 ft	PoE / PoE++ from Flex switch
Center pole	U6 Mesh Pro #2	Radiate over C1/C2; reaches P1	~15-18 ft	PoE from Flex switch
Court 3 pole	Link B Bridge RX	Aim at center pole	~18-20 ft	PoE from Court 3 switch
Court 3 pole	Small PoE switch	In NEMA enclosure on pole	~6 ft (reach)	AC from pole light circuit
Court 3 pole	G6 Bullet (Court 3)	Aim down length of Court 3	~18-20 ft	PoE from Court 3 switch
Court 3 pole	U6 Mesh Pro #3	Radiate over Court 3	~15-18 ft	PoE from Court 3 switch
Parking lot (existing)	G6 180	No change - keep on lot	existing	existing
Parking lot (existing)	AI PTZ Industrial	No change - STAYS on lot (not moved)	existing	existing

Camera aiming: mount each bullet at one END of its court and aim down the 60 ft length for full-court coverage and best pixel density. An OPTIONAL action PTZ on the center pole provides zoom/follow for special games only (operated on demand). Do not use the G6 180 for court detail. The AI PTZ Industrial remains on the parking lot.

5. Bill of Materials

New hardware to purchase

Item	Qty	Unit (approx)	Line (approx)
Wireless bridge link (UBB pair, or airMAX pair)	2	\$200-360	\$400-720
UniFi Protect G6 Bullet (4K, IP66)	3	\$199	\$597
Small PoE switch - Court 3 (Flex Mini / Flex)	1	\$30-65	\$30-65
Outdoor NEMA enclosure (center pole + Court 3)	2	\$79	\$158
Ethernet surge protector (ETH-SP)	4	\$25	\$100
Building bridge mast + pole mounts + patch cables	1 lot	~\$120	~\$120

Estimated net new spend (excl. action cam): approx. \$1,400 - \$1,900

Action camera for special games - CHOOSE ONE (optional)

- Repurpose a SPARE owned PTZ (G6/G5 PTZ from existing fleet) - \$0 (preferred if available).
- Buy a new UniFi G6 PTZ (4K, optical zoom; UniFi Protect native) - approx. \$1,599.
- Portable streaming cam (Mevo Start / GoPro on tripod) for highlights/livestream - approx. \$300-500 (not UniFi-integrated; deployed only for special games).

Mounts on the center pole and operated on demand only; otherwise idle. The center-pole switch/PoE is provisioned for it either way.

Reused (already owned - \$0)

- 3x U6 Mesh Pro APs - relocated off building wall to poles
- Switch Flex 2.5G PoE + 210W adapter - center-pole local switch
- Network Video Recorder Pro + 16 TB HDD - recording
- AI PTZ Industrial - stays on parking lot (not moved)
- Spare PoE++ 60W injector, SFP-to-RJ45 adapters, DAC cables

Consumables / verify on hand

- Outdoor/UV-rated or direct-burial shielded Cat6 (short patch runs at each pole)
- Stainless pole-mount brackets / band clamps sized for 20 ft round poles
- Weatherproof cable glands, UV zip ties, drip loops, silicone/dielectric grease
- RJ45 outdoor connectors + tester

6. Installation & Configuration

Physical install sequence

1. Mount Link A bridge on building roofline (above parapet); run PoE patch to a rack switch port.
2. Mount the NEMA enclosure on the center pole; install the Flex 2.5G PoE switch + 210W adapter; have an electrician land pole-light-circuit power to the enclosure (no trenching - power is at pole).
3. Mount center-pole devices: Link A RX, Link B TX, 2x G6 Bullet, action PTZ (if fitted), U6 Mesh Pro #2. Patch all to the Flex switch.
4. Mount Court 3 enclosure + small PoE switch; land pole power; mount Link B RX, G6 Bullet, U6 Mesh Pro #3.
5. Relocate the U6 Mesh Pro from the building wall to the P2 post; feed via the existing pulled cable (land it on a rack PoE port).
6. Install ETH-SP surge protectors at each outdoor cable end (bridge feeds and switch uplinks).
7. Aim both bridge links; aim cameras down each court; tilt APs slightly down toward the sand.

UniFi configuration

- Adopt both bridge radios; set as transparent L2 bridge (UBB is automatic; airMAX = bridge mode).
- Adopt the relocated APs; build one outdoor WLAN; enable band steering. Set fixed, non-overlapping channels (5 GHz) across the 3 APs to avoid co-channel interference.
- Reduce AP TX power from Auto/High to Medium once placed on poles - prevents the APs from over-talking each other in the open field; verify with a follow-up heat map.
- Adopt cameras in UniFi Protect; confirm they record to the NVR Pro; set motion/AI zones to the court areas; create PTZ presets for Court 1 and Court 2.
- If an action PTZ is fitted: high-end PTZs draw PoE++ - land it on a PoE++ capable port of the Flex switch (210W budget) or inline the spare 60W PoE++ injector.

PoE budget - center pole Flex 2.5G PoE switch

Load on center-pole switch	Approx draw
2x G6 Bullet	~14 W
Action PTZ if fitted (PoE/PoE++)	~10-45 W
U6 Mesh Pro #2	~13 W
Link A RX + Link B TX bridges	~14 W
TOTAL (plan for headroom)	~55-90 W

The 210W adapter provides ample budget; just confirm the PTZ lands on a PoE++ port.

7. Commissioning & Acceptance Checklist

Wireless backhaul

- ☐ Link A connected; UniFi shows link rate and signal (target RSSI better than -60 dBm).
- ☐ Link B connected; Court 3 devices online through the relay.
- ☐ Failover sanity: unplug/replug - links auto-recover.

WiFi coverage (re-survey)

- ☐ Re-run the WiFiman heat map walking all 3 courts + both patios.
- ☐ Court surface now reads -55 dBm or better (was -70 to -80).
- ☐ No dead pocket on P1 by the building (add/keep an AP there if weak - wired drops exist).
- ☐ Roaming: a phone walks court-to-court without dropping.

Cameras (UniFi Protect)

- ☐ All 3 fixed court bullets (+ action PTZ if fitted) online and recording to NVR Pro.
- ☐ Each bullet frames its full court end-to-end; if PTZ fitted, presets for C1/C2 saved.
- ☐ Night check: image usable under the Musco court lights.
- ☐ Retention check: 16 TB HDD holds target days with the added cameras (add HDD if short).

Power & weatherproofing

- ☐ Enclosures sealed; cable glands tight; drip loops on all outdoor cables.
- ☐ ETH-SP surge protectors grounded at each outdoor run.
- ☐ PTZ confirmed on PoE++; no under-power warnings in UniFi.

Sign-off

Installer: _____

Date: _____

Notes: _____
